

Spectral-Reset Cross-Action Cancellation

An Exact Tail-Coupling Identity and a Four-Mode Bridge Obstruction

Prime Dynamics Theory Program

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Abstract

RH-156 closes a finite native reset support floor, but the earlier directional architecture uses a different object: the projected recent cross $(I - P)RP$. We determine whether a contemporaneous full-memory spectral reset can bridge the two constructions.

Let $M = R + T$ and let P be a spectral projector of M . Spectral invariance gives the exact cancellation

$$(I - P)RP = -(I - P)TP.$$

Thus the recent projected cross sees only old-tail coupling. A positive and well-conditioned native compression PRP supplies no positive lower for any cross singular value; the obstruction remains even when $\|T\|$ is fixed and nonzero, because T may commute with P .

We derive an outward perturbation radius for $J(P, T) = (I - P)TP$. The frozen tail is exactly $T_t = \eta^d M_{t-d}$, so its matrix radius is inherited from RH-151. Fifty of 130 snapshots have inactive tail and therefore exact zero cross action. Of the 80 active snapshots, only 54 certify a positive fourth coupling singular value; 26 hit the packet/tail radius wall. The RH-154 terminal half certifies only 30 of 62 snapshots, and only two of ten terminal snapshots have four certified modes. Hence the native support cannot be identified directly with the old projected-cross support. A lagged or hybrid reset packet is the next controlled alternative.

1 Two support objects that need not agree

The native pair uses PRP and PTP inside the selected packet. The older directional route instead starts from the off-diagonal action

$$K_R = (I - P)RP,$$

selects four right singular directions, and builds a wedge/capacity support from their images. Positivity of PRP does not normally imply that K_R has any rank. For a spectral reset the relation is even more rigid.

2 Exact cross-action cancellation

Theorem 1 (Spectral-reset cancellation). *Let $M = R + T$ be Hermitian and let P be any spectral projector of M . Then*

$$(I - P)RP = -(I - P)TP.$$

Consequently

$$\sigma_k((I - P)RP) = \sigma_k((I - P)TP) \leq \|T\|$$

for every singular-value index k .

Proof. Spectral invariance gives $(I - P)MP = 0$. Substitute $M = R + T$ and rearrange. The singular-value identity follows by multiplication by -1 , and the upper bound follows from contractivity of orthogonal projections. $\square$

Proposition 2 (No native-to-cross lower). *No positive lower for $\sigma_k((I - P)RP)$ follows from positive eigenvalue endpoints for PRP , a positive overlap lower, and an upper or exact value of $\|T\|$. This remains true for every $k \geq 1$.*

Proof. Choose R, T , and P diagonal in one common basis, with $PRP \succ 0$ and $\|T\|$ equal to any prescribed nonnegative value. Then $(I - P)TP = 0$, so Theorem 1 gives zero recent cross. The native Gram can simultaneously have arbitrary positive endpoint bounds. $\square$

The missing datum is therefore explicit: a lower bound on the off-diagonal tail coupling itself, such as $\sigma_4((I - P)TP) \geq j_* > 0$. Tail mass alone points in the opposite direction by supplying only an upper.

3 Outward tail-coupling balls

Let Q be the exact reset projector, P its center, $\|Q - P\| \leq \epsilon$, and let the exact tail obey $\|T - \hat{T}\| \leq r_T$.

Theorem 3 (Coupling perturbation radius). *For $J(P, T) = (I - P)TP$,*

$$\|J(Q, T) - J(P, \hat{T})\| \leq r_T + 2\|\hat{T}\|\epsilon.$$

Hence each exact coupling singular value lies within this radius of its nominal counterpart.

Proof. First change the tail at fixed Q , costing at most r_T . At fixed $\hat{T}$, expand

$$J(Q, \hat{T}) - J(P, \hat{T}) = (P - Q)\hat{T}Q + (I - P)\hat{T}(Q - P).$$

The two terms cost at most $\|\hat{T}\|\epsilon$ each. Singular-value Weyl bounds finish the proof [Bhatia, 1997, Stewart and Sun, 1990]. $\square$

For geometric memory at depth d ,

$$T_t = \eta^d M_{t-d} \quad (t \geq d).$$

Therefore the archived RH-151 full-memory radius at time $t - d$, multiplied by η^d , is already a tail radius. No new source interval computation is required.

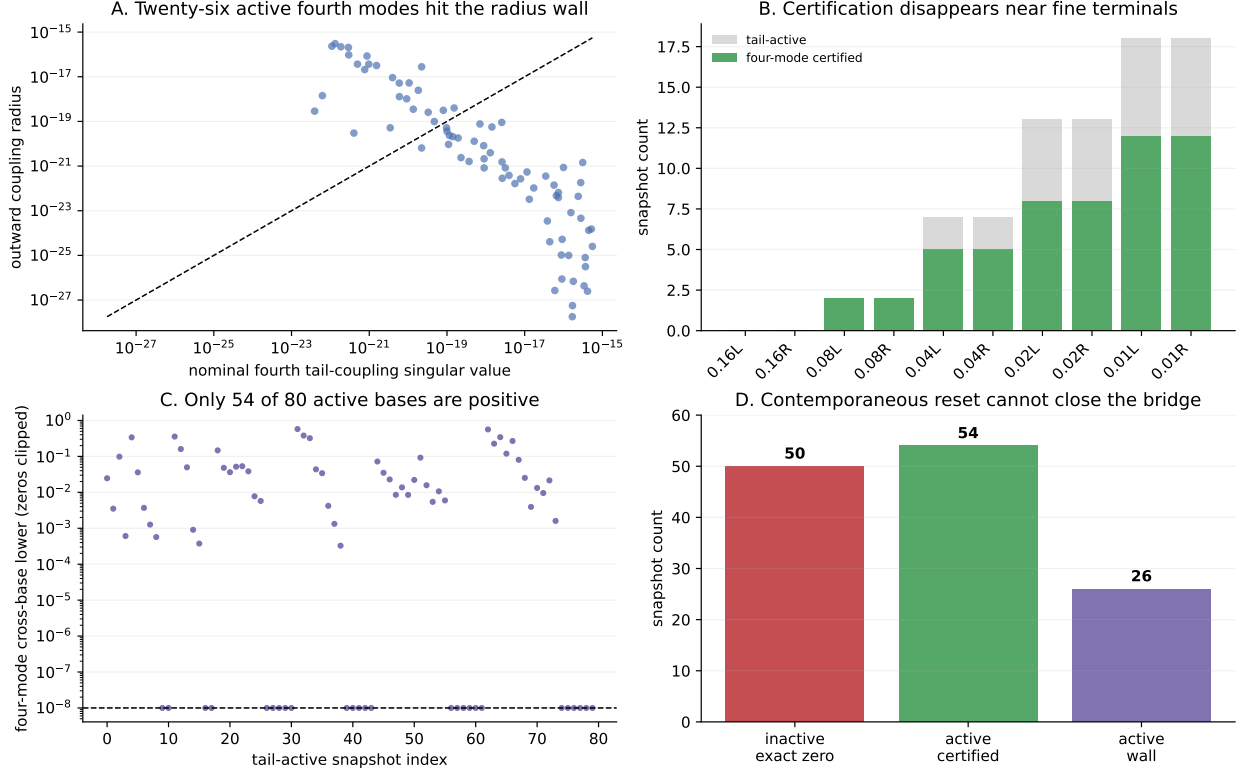


Figure 1: Fourth-coupling radii, channel counts, positive normalized bases, and the exact-zero/positive/wall decomposition.

4 The 130-snapshot bridge audit

At the fifty snapshots with $t < d = 5$, the exact tail is zero. Theorem 1 makes the exact recent cross zero as well, regardless of the strong native support floor.

The remaining eighty snapshots have active tail. Applying Theorem 3 to the fourth singular value certifies 54 positive four-mode couplings and leaves 26 at zero lower. Positive normalized four-mode base lowers range from 3.2783×10^{-4} to 0.57658, with median 0.02495. Their absolute fourth-singular lowers are tiny, from 1.58×10^{-20} to 5.53×10^{-16} , reflecting the η^5 tail scale.

Only the two $\sigma = 0.08$ channels certify every active snapshot. At finer scales the fourth-mode lower disappears near the terminal weak-gap region. Only two of ten terminal snapshots certify four modes. The delayed terminal-half atlas contains 62 snapshots, six with exact zero tail and only 30 with positive fourth-coupling lower. Delaying removes overlap births but cannot repair spectral invariance.

5 Route consequence and boundary

RH-157 rejects one tempting identification without damaging RH-156. The native reset support is a valid finite functional, but the contemporaneous spectral packet is deliberately

invariant under the full memory; its projected recent cross is therefore only a small tail leakage. Forcing it into the recursive directional architecture discards the main recent energy.

A lagged packet P_{t-1} is not invariant under M_t and may restore a substantial cross while RH-152 controls its angle to P_t . A hybrid between P_{t-1} and P_t is another controlled option. We have not proved either bridge, an all-level four-coupling law, Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.

G. W. Stewart and Ji-Guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.